KTH Computer Science and Communication

Exam in DD2421 Machine Learning
2025-03-12, kl 8.00 – 12.00

Aids allowed: *calculator*, *language dictionary*.

To take this exam you must be registered to this specific exam as well as to the course.

In order to pass this exam, your score x first needs to be 20 or more (out of 42, full point). In addition, given your points y from the Programming Challenge (out of 18, full point), the requirements on the total points, $p = x + y$, are preliminarily set for different grades as:

$$
\begin{aligned}
54 < p \leq 60 &\quad \rightarrow \quad A \\
48 < p \leq 54 &\quad \rightarrow \quad B \\
42 < p \leq 48 &\quad \rightarrow \quad C \\
36 < p \leq 42 &\quad \rightarrow \quad D \\
29 < p \leq 36 &\quad \rightarrow \quad E \quad \text{(A pass is guaranteed with the required points for 'E'.)} \\
0 \leq p \leq 29 &\quad \rightarrow \quad F
\end{aligned}
$$

This exam consists of sections **A**, **B**, and **C**.
NB: use different papers (answer sheets) for different sections A, B, and C.
This means you shall start a new page after section A, and again after section B.

A Graded problems

Potential inquiries to be addressed to Atsuto Maki.

A-1 Terminology (5p)

For each term (a–e) in the left list, find the explanation from the right list which *best* describes how the term is used in machine learning.

a) Occams's razor

b) k-fold cross validation

c) OOD examples

d) RANSAC

e) Dropout

1) Training samples representing a class

2) An approach to fit a model to data with outliers

3) A technique for margin maximization

4) Algorithm to learn with latent variables

5) A technique used in train artificial neural networks

6) Robust method for Monte Carlo tree search

7) A technique for assessing a model by exploiting available data for training and testing

8) A principle to choose the simplest explanation

9) An approach to find useful dimension for classification

10) To be rejected as "don't know" category

Solution: a-8, b-7, c-10, d-2, e-5

A-2 Nearest Neighbor, Classification (4p)

Suppose that we take a data set, divide it into training and test sets, and then try out two different classification procedures. We use *three-quarters* of the data for training, and the remaining *one-quarter* for testing. First we use *Bagging* and get an error rate of 10% on the training data. We also get the average error rate (weighted average over both test and training data sets) of 12%. Next we use 1-nearest neighbor and get an average error rate (weighted average over both test and training data sets) of 6%.

a) What was the error rate with 1-nearest neighbor on the training set? (1p)

b) What was the error rate with 1-nearest neighbor on the test set? (1p)

c) What was the error rate with Bagging on the test set? (1p)

d) Based on these results, indicate the method which we should prefer to use for classification of new observations, with a simple reasoning. (1p)

Solution:

a) 0%. Training error for 1-NN is always zero.
b) 24%. Given the answer in **a**), the testing error is 24%.
c) 18%.
d) Bagging, because it achieves lower error rate on the test data (18% < 24%).

A-3 Entropy and Decision Trees/Forests (5p)

a) Consider a single toss of *skewed* coin (it is likely to show one side more than the other side). Regarding the uncertainty of the outcome {head, tail}, which one of the following is correct?

i. the entropy is smaller than one bit.
ii. the entropy is equal to one bit.
iii. the entropy is greater than two bits.

Motivate your answer by short phrases. (1p)

b) Indicate any valid ones as the basic strategy for selecting a question (attribute) at each node in decision trees.

i. To maximize the expected reduction of the entropy.
ii. To minimize the expected reduction of the entropy.
iii. To minimize the expected reduction of gini impurity.

Motivate your answer by short phrases. (1p)

c) How is Shannon's information content defined for an event i with probability $P(i)$? (1p)

d) Suppose we have trained a Decision Forest using five bootstrapped samples from a data set containing three classes, {Red, Blue, Green}. We then applied the model to a specific test input, $\mathbf{x}$, and observed five estimates of class distributions from the five trees, respectively:

$\{P_1(\text{Red}|\mathbf{x}), P_1(\text{Blue}|\mathbf{x}), P_1(\text{Green}|\mathbf{x})\}$ = {0.1, 0.6, 0.3}
$\{P_2(\text{Red}|\mathbf{x}), P_2(\text{Blue}|\mathbf{x}), P_2(\text{Green}|\mathbf{x})\}$ = {0.2, 0.3, 0.5}
$\{P_3(\text{Red}|\mathbf{x}), P_3(\text{Blue}|\mathbf{x}), P_3(\text{Green}|\mathbf{x})\}$ = {0.3, 0.5, 0.2}
$\{P_4(\text{Red}|\mathbf{x}), P_4(\text{Blue}|\mathbf{x}), P_4(\text{Green}|\mathbf{x})\}$ = {0.2, 0.3, 0.5}
$\{P_5(\text{Red}|\mathbf{x}), P_5(\text{Blue}|\mathbf{x}), P_5(\text{Green}|\mathbf{x})\}$ = {0.2, 0.4, 0.4}

Let us consider the average probability for combining these results into a single class prediction. What will the final class prediction be? *Motivate your answer by short phrases*. (1p)

e) What is the kind of additional randomness involved in the design of Decision Forests with respect to Bagging? (1p)

Solution:
a) i.
b) i.
c) $log_2 1/P(i)$
d) Blue, (0.2,0.42,0.38)
e) Randomness in possible features to use at each node.

A-4 Regression with regularization (4p)

a) In regression, regularization is a process of introducing additional term, so-called *shrinkage penalty*. Which ones of the three methods includes the additional term.

i. LASSO.
ii. Logistic regression.

iii. k-NN regression.

Simiply indicate your choice. (1p)

b) For a set of N training samples $\{(\mathbf{x}_1, y_1), \ldots, (\mathbf{x}_N, y_N)\}$, each consisting of input vector $\mathbf{x}$ and output y, suppose we estimate the regression coefficients $\mathbf{w}^\top (\in \mathbf{R}^d) = \{w_1, \ldots, w_d\}$ in a linear regression model by minimizing

$$\sum_{n=1}^{N} (y_n - \mathbf{w}^\top \mathbf{x}_n)^2 + \lambda \sum_{i=1}^{d} w_i^2$$

for a particular value of λ.

Now, let us consider different models trained with different values of λ, starting from a very large value (infinity) and *decreasing* it down to 0. Then, the training error and test error (residual sum of squares, RSS) are believed to change. Indicate which of i. through v. is most likely to be the case. *Justify your answer.* (2p)

i. Remain constant.
ii. Steadily increase.
iii. Steadily decrease.
iv. Decrease initially, and then eventually start increasing in a U shape.
v. Increase initially, and then eventually start decreasing in an inverted U shape.

c) As we *decrease* λ, from infinity, what happens to the variance of the model. Choose the answer from the five alternatives above, and *briefly justify your answer.* (1p)

Solution: a)-i, **b**)-iii (training error), **b**)-iv (testing error), **c**)-ii
If λ was infitity, all w_i would be zero; the model is extremely simple, predicting a constant and has no variance with the prediction being far from actual value (thus with high bias). As we decrease λ, all w_i increase from zero toward their least square estimate values while steadily decreasing the *training* error to the ordinary least square RSS (and also decreasing bias) as the model continues to better fit training data. The values of w_i then become more dependent on training data, thus increasing the variance. The *test* error initially decreases as well, but eventually start increasing due to overfitting to the training data.

A-5 Dimensionality Reduction, Subspace Methods (4p)

We consider to solve a K-class classification problem with the Subspace Method and for that we compute a subspace $\mathcal{L}^{(j)}$ $(j = 1, ..., K)$ using training data for each class, respectively. That is, given a set of feature vectors (as training data) which belong to a specific class C (i.e. with an identical class label), we perform PCA on them and generate an orthonormal basis $\{\mathbf{u}_1, ..., \mathbf{u}_p\}$ which spans a p-dimensional subsapce, $\mathcal{L}$, as the outcome.

Provide an answer to the following questions.

a) Given that we compute $\{\mathbf{u}_1, ..., \mathbf{u}_p\}$ as eigenvectors of the auto-correlation matrix Q based on the training data, how should we choose the eigenvectors in relation to the corresponding eigenvalues of Q? (1p)

b) We have a new input vector $\mathbf{x}$ whose class is unknown, and consider its projectiton length on $\mathcal{L}^{(j)}$ $(j = 1, ..., K)$. Briefly explain the way to determine the class to which vector $\mathbf{x}$ should belong using the projection lengths. (1p)

c) Now, describe how the projection length is represented, using a simple formula. (2p)

Solution:

a) Choose the eigenvectors corresponding to the largest eigenvalues of Q.

b) $\mathbf{x}$ should belong to the class where the projection length to the corresponding subspace is the maximum.

c) $\sqrt{S}$ where $S = \Sigma_{i=1}^{p}(\mathbf{x}, \mathbf{u}_i)^2$

B Graded problems

Potential inquiries to be addressed to Olov Andersson.

B-1 Probability Theory (4p)

a) Assume that you have a joint probability distribution $P(A, B)$. First *write* the formula for Bayes' theorem and then *show how to derive* it. (2p)

b) Assume you have five random variables A,B,C,D,E. For each of the following expressions 1 to 4 below, state if they are *equivalent* or *not equivalent* to the joint probability distribution $P(A, B, C, D)$:

1) $P(A, B, C|D)P(D)$
2) $P(A|B)P(B|C)P(C, D)$
3) $P(C, D|A, B)P(A, B)$
4) $\int_E P(A, B, C, D, E)$ (2p)

Solution:

a) $P(A|B) = P(B|A)P(A)/P(B)$. This can be derived via the definition of conditional probability or two factorizations of the same joint distribution into both $P(A|B)(B)$ and $P(B|A)P(A)$, and just rearranging the terms (see Lecture 5 slides for both).

b) 1) Eqv. (valid factorization)
2) Not eqv.
3) Eqv. (reordering and factorization)
4) Eqv. (marginalization rule)

B-2 Maximum likelihood estimation for Regression (4p)

You wish to model a labeled dataset $\mathcal{D} = ((x, y)_i : x, y \in \mathbb{R})$ of N observations with the following model:

$$y = w_0 + w_1\, sin(2x) + \epsilon \tag{1}$$

where $\epsilon \sim \mathcal{N}(0, \sigma^2)$, i.e.,

$$P(\epsilon) = \frac{1}{\sqrt{2\pi\sigma^2}} e^{-\epsilon^2/2\sigma^2}. \tag{2}$$

Derive a formula for the maximum likelihood estimate of w_0 as a function of w_1. *Show the steps of your derivations*.

Solution: The random variable $Y|X = x$ is distributed

$$Y|X = x \sim \mathcal{N}(w_0 + w_1\, sin(2x), \sigma^2). \tag{3}$$

The maximum likelihood optimality criterion says

$$
\begin{aligned}
(w_0, w_1)_{ML} &= \arg\max_{w_0,w_1} Pr(\mathcal{D}|w_0, w_1) \qquad (4)\\
&= \arg\max_{w_0,w_1} \sum_i \log Pr(y_i|X = x_i)\\
&= \arg\max_{w_0,w_1} \sum_i \log \frac{1}{\sqrt{2\pi\sigma^2}} e^{-(y_i-(w_0+w_1\, sin(2x)))^2/2\sigma^2}\\
&= \arg\max_{w_0,w_1} \sum_i -\log\sqrt{2\pi\sigma^2} - (y_i - (w_0 + w_1\, sin(2x)))^2/2\sigma^2\\
&= \arg\min_{w_0,w_1} \sum_i (y_i - (w_0 + w_1\, sin(2x)))^2. \qquad (5)
\end{aligned}
$$

Taking the partial derivative of this expression with respect to w_0 produces:

$$
\frac{\partial}{\partial w_0} \sum_i (y_i - (w_0 + w_1\, sin(2x)))^2 = \sum_i 2(y_i - (w_0 + w_1\, sin(2x)))(-1). \qquad (6)
$$

Setting to zero and solving for w_0 gives:

$$
w_0(w_1) = \frac{1}{N} \sum_i (y_i - w_1\, sin(2x)). \qquad (7)
$$

B-3 Maximum likelihood estimation for Classification (4p)

Consider a binary classification problem with only two data points (for simplicity), where the output class y_i is encoded as 0 or 1:

$$
\mathcal{D} = ((x_1, y_1), (x_2, y_2)) = ((1, 0), (2, 1))
$$

You decide to fit a simple logistic regression model with parameter θ, defined by:

$$
P(y = 1 \mid x, \theta) = \sigma(\theta x) = \frac{1}{1 + e^{-\theta x}},
$$

where $\sigma(\cdot)$ is the logistic sigmoid function.

a) *Write* the likelihood function $L(\theta)$ with the numbers from the given data set. (2p)

b) Show how one can one find the maximum likelihood estimate of θ (you do not need to calculate the value of θ, but concisely explain how to do that). (2p)

Solution:

a) The likelihood function $L(\theta)$ for logistic regression is given by:

$$
L(\theta) = P(y_1 = 0 \mid x_1, \theta) \cdot P(y_2 = 1 \mid x_2, \theta).
$$

Given our data:

$$
L(\theta) = [1 - \sigma(\theta \cdot 1)] \cdot [\sigma(\theta \cdot 2)]\,.
$$

Substituting the sigmoid definition explicitly, we have:

$$
L(\theta) = \left[1 - \frac{1}{1 + e^{-\theta}}\right] \cdot \frac{1}{1 + e^{-2\theta}} = \frac{e^{-\theta}}{1 + e^{-\theta}} \cdot \frac{1}{1 + e^{-2\theta}}.
$$

b) The log-likelihood function $\ell(\theta) = \log L(\theta)$ is:

$$\ell(\theta) = \log\left(\frac{e^{-\theta}}{1+e^{-\theta}}\right) + \log\left(\frac{1}{1+e^{-2\theta}}\right).$$

Simplifying further, we get:

$$\ell(\theta) = -\theta - \log(1+e^{-\theta}) - \log(1+e^{-2\theta}).$$

To find the MLE, we differentiate $\ell(\theta)$ with respect to θ and can attempt to set the derivative equal to zero to solve the equation:

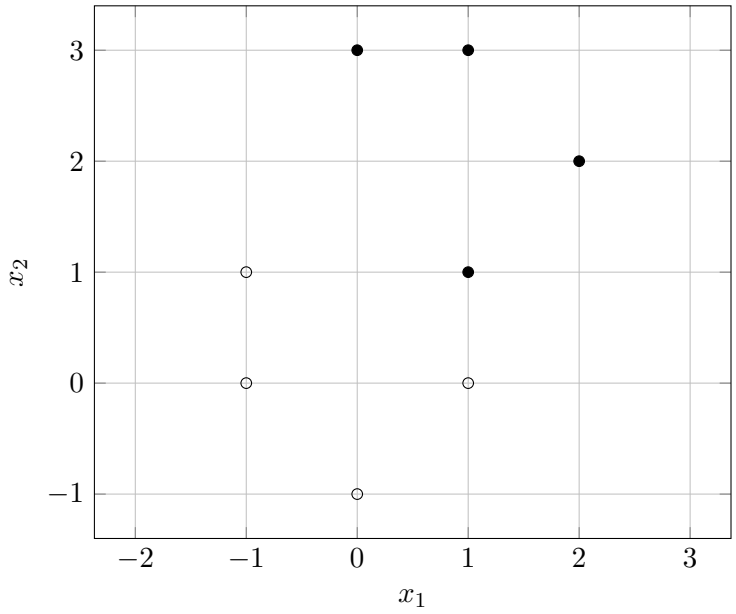
$$\frac{d\ell(\theta)}{d\theta} = -1 + \frac{e^{-\theta}}{1+e^{-\theta}} + \frac{2e^{-2\theta}}{1+e^{-2\theta}} = 0.$$

However, this lacks a closed-form solution. The maximum of θ in logistic regression is instead typically found via numerical optimization (e.g. gradient descent, Newton's method etc).

C Graded problems

Potential inquiries to be addressed to Jörg Conradt.

C-1 Support Vector Machines (5p)



The eight datapoints in the figure are used to train a support vector machine. Filled circles are positive samples and unfilled circles are negative samples.

When a linear kernel ($\mathcal{K}(\vec{x}, \vec{y}) = (\vec{x} \cdot \vec{y} + 1)$) was used, and no slack was allowed ($C = \infty$), this resulted in three support vectors, with the corresponding α-values 0.5, 2, and 2.5.

To classify new points, the normal indicator function is used:

$$\text{ind}(\vec{x}) = \sum_k t_k \alpha_k \mathcal{K}(\vec{x}, \vec{s}_k) - b$$

a) What will the three support vectors be? You do not need to calculate this mathematically; use you knowledge about support vectors to deduce which three points must be support vectors. (2p)

b) Which α-value must be associated with each of the three support vectors? (Give short motivations, ideally keywords) (2p)

c) What value for b should be used in the indicator function? (1p)

Solution:

a)

$$s_1 = \begin{bmatrix} -1 \\ 1 \end{bmatrix} \qquad s_2 = \begin{bmatrix} 1 \\ 0 \end{bmatrix} \qquad s_3 = \begin{bmatrix} 1 \\ 1 \end{bmatrix}$$

The decision boundary has to be a straight line, passing between the two classes. Removing any one of these three datapoints would clearly change the position of the decision boundary. Therefore, they must all be support vectors, since these are the points that define the boundary.

b) $\sum_k t_k \alpha_k = 0$ means that the single positive support vector (s_3) must have the highest α-value. Of the other two, the point closest to the opposite class (s_2) will have the larger α. Therefore:

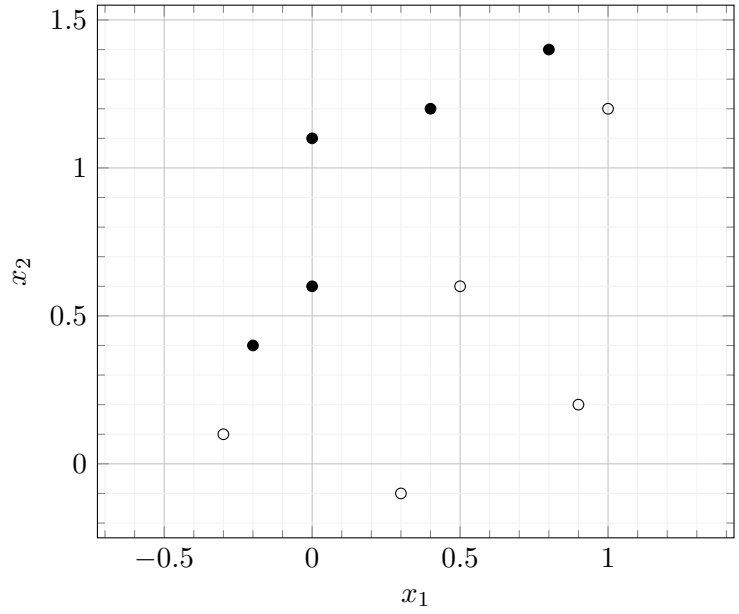
$$\alpha_1 = 0.5 \qquad \alpha_2 = 2.0 \qquad \alpha_3 = 2.5$$

c) The indicator function value is known to be 1 or -1 on the margin border, where the support vectors are (since we have no slack).

Using $\text{ind}(s_1) = -1$ (this is a negative point) we can calculate b as

$$b = \sum_k t_k \alpha_k \mathcal{K}(s_1, s_k) + 1 = -0.5 \cdot 3 + 2.0 \cdot 0 + 2.5 \cdot 1 + 1 = 2.0$$

C-2 Linear classifier, single neuron (3p)



a) The figure illustrates a dataset where filled circles are positive samples and unfilled are negative. For a single "artificial neuron", find a set of values for the weights and the threshold that will separate the positive from the negative samples. Make sure that positive points are *above* the threshold and negative points are *below*.

Give exactly ONE equation that shows the output of the neuron. Ensure that your equation contains all input variables, all weights, and a bias (=threshold) value. (2p)

Note: There is no need to use any learning algorithm here. It is sufficient to find a suitable linear separator in the figure, but you must explicitly state the weight and threshold values and explain how you arrived at these values IN KEYWORDS or short sentences.

b) If a new negative sample at $(0, 0)$ arrives, and learning is done with the *perceptron learning rule*; how will these values change? (1p)

Solution:

a) A natural separator is the line passing through $(-0.5, 0.0)$ and $(1.0, 1.5)$. The weight vector is always perpendicular to this line and points towards the positive sample side (here: upper left).

$$w_1 = -1, \qquad w_2 = 1$$

We get the threshold, θ, by testing a point on the line, for example $(0.0, 0.5)$, using the weights we have chosen.

$$0.0 \cdot w_1 + 0.5 \cdot w_2 - \theta = 0 \quad \Rightarrow \quad \theta = 0.5$$

b) No change, because the sample is already correctly classified.